

Top 80 TMUA Mistakes

A tight, student-facing mistake bank for TMUA. The aim is not just to read this: tag every wrong answer after timed practice, find your repeat errors, and drill the exact fix.

How to use this: after every timed paper, label each wrong answer with one mistake number. If the same mistake appears in 3 sittings, stop doing random practice and drill that one error type.

Domain

Zero, negatives, logs,
square roots,
denominators.

Sign

Inequalities, indices,
brackets, trig quadrants.

Graph

Transformations,
intersections, area,
sketching.

Logic

If, only if, converse,
contrapositive,
quantifiers.

MCQ

Trap answers, edge
cases, timing,
elimination.

The Big 10 - check these first

Rank	Mistake	Quick exam check
1	Dividing by a variable and deleting the zero case	Could the thing I cancelled be zero?
2	Forgetting $\sqrt{x^2} = x $	Did a square root hide a sign?
3	Not flipping an inequality after multiplying/dividing by a negative	Did I divide by something negative or possibly negative?
4	Wrong Remainder Theorem substitution	Did I set the divisor equal to zero?
5	Confusing converse and contrapositive	Which arrow is actually being used?
6	Wrong negation of "if A, then B"	Negation is A true and B false.
7	Missing graph transformation reversal	$f(x + a)$ moves left, not right.
8	Confusing signed integral with area	Did the graph cross the axis?
9	Missing trig solutions in a given interval	Did I check all quadrants/periods?
10	Using examples as proof	Does this prove every case or just test some cases?

1. Timed paper

75 minutes strict. No pausing.

2. Tag errors

Use mistake numbers, not
vague notes.

3. Drill repeats

Same error 3 times = targeted
practice.

4. Re-test

Target: fewer repeat mistakes
next sitting.

1-10. Algebra and domain traps

1 Dividing by a variable without checking zero

P1 HIGH COST

TRAP: From $x^2 = 5x$, cancelling x gives $x = 5$ and loses $x = 0$.

FIX: Move all terms to one side and factorise: $x^2 - 5x = x(x - 5) = 0$.

2 Forgetting $\sqrt{x^2} = |x|$, not x

P1 HIGH COST

TRAP: Writing $\sqrt{(x - 3)^2} = x - 3$ for all x .

FIX: Use $\sqrt{A^2} = |A|$. Then split into cases if needed.

3 Not flipping inequalities after negatives

P1 HIGH COST

TRAP: Solving $-2x > 6$ as $x > -3$.

FIX: Dividing by a negative flips the sign: $x < -3$.

4 Multiplying an inequality by unknown sign

P1

TRAP: From $1/x > 2$, multiplying by x without knowing whether x is positive or negative.

FIX: Bring to one side or split cases: $(1 - 2x)/x > 0$.

5 Splitting square roots over addition

P1

TRAP: Using $\sqrt{a + b} = \sqrt{a} + \sqrt{b}$.

FIX: Roots distribute over products, not sums: $\sqrt{ab} = \sqrt{a}\sqrt{b}$, but $\sqrt{a+b}$ does not split.

6 Forgetting excluded values in rational expressions

P1

TRAP: Cancelling $x(x - 2)/x$ and then allowing $x = 0$.

FIX: Before cancelling, write restrictions: denominator $\neq 0$.

7 Solving after squaring without checking answers

P1

TRAP: Squaring both sides creates an extra solution that still appears in the answer options.

FIX: Any time you square both sides, substitute final answers back into the original equation.

8 Ignoring log and root domains

P1 HIGH COST

TRAP: Solving $\log(x - 2)$ while allowing $x \leq 2$.

FIX: Write domain first: log input > 0 ; even-root input ≥ 0 ; denominator $\neq 0$.

9 Forgetting plus/minus after square-rooting

P1 HIGH COST

TRAP: From $x^2 = 9$, writing $x = 3$ only.

FIX: Square-rooting gives two cases: $x = \pm 3$, unless the context restricts x .

10 Losing equality in inequality problems

P1

TRAP: Solving $x^2 \leq 9$ but giving $-3 < x < 3$.

FIX: Track strict vs non-strict signs. Roots are included for $\leq$ or $\geq$ when defined.

11-20. Indices, quadratics, logs and sequences

11 Confusing index laws

P1 HIGH COST

TRAP: Using $a^m a^n = a^{mn}$ or $(a^m)^n = a^{m+n}$.

FIX: Same-base multiply means add powers. Power of a power means multiply powers.

12 Negative index sign confusion

P1

TRAP: Thinking $a^{-2} = -a^2$.

FIX: A negative power means reciprocal: $a^{-2} = 1/a^2$.

13 Wrong discriminant or wrong conclusion

P1

TRAP: Using $b^2 + 4ac$, or saying $\Delta < 0$ gives two real roots.

FIX: $\Delta = b^2 - 4ac$. $\Delta > 0$: two real; $= 0$: repeated; < 0 : no real roots.

14 Completing the square with the wrong sign

P1

TRAP: Writing $x^2 + 6x + 5 = (x + 3)^2 + 5$.

FIX: Compensate outside: $x^2 + 6x + 5 = (x + 3)^2 - 4$.

15 Using log laws without checking positivity

P1

TRAP: Accepting $x = 1$ in $\log x + \log(x - 3)$.

FIX: Before log laws, every log input must be positive.

16 Wrong substitution in factor/remainder theorem

P1 HIGH COST

TRAP: For divisor $2x - 3$, substituting $x = 3$ or $x = -3$.

FIX: Set the divisor equal to zero. For $2x - 3$, use $x = 3/2$, so remainder is $f(3/2)$.

17 Finite vs infinite geometric series confusion

P1 HIGH COST

TRAP: Using $S_{\infty} = a/(1 - r)$ for a finite series, or when $|r| > 1$.

FIX: Finite: $S_n = a(1 - r^n)/(1 - r)$. Infinite only if $|r| < 1$.

18 Wrong sign or term in binomial expansion

P1 HIGH COST

TRAP: Expanding $(1 - 2x)^5$ and making every term positive.

FIX: Keep the sign inside the term: $C(n, r)a^{n-r}b^r$. Odd powers of $-2x$ stay negative.

19 Quadratic sequence coefficient error

P1

TRAP: Thinking the second difference equals a in $an^2 + bn + c$.

FIX: For $u_n = an^2 + bn + c$, the second difference is $2a$.

20 Solving simultaneous equations but not back-substituting

P1

TRAP: Finding possible x -values but selecting an option before finding the matching y -values.

FIX: After solving the reduced equation, substitute back and check each ordered pair.

21-30. Functions, graphs and coordinate geometry

21 Graph transformation reversal

P1 HIGH COST

TRAP: Thinking $f(x + 3)$ shifts right by 3.

FIX: Inside transformations are opposite: $f(x + 3)$ moves left 3. $f(ax)$ has horizontal factor $1/a$.

22 Confusing $af(x)$ with $f(ax)$

P1

TRAP: Treating vertical and horizontal stretches as the same.

FIX: Outside affects y-values. Inside affects x-values in the opposite direction.

23 Forgetting function order in composition

P1

TRAP: Thinking $f(g(x))$ means do f first.

FIX: Do g first, then feed the result into f.

24 Intersections: solving only one graph

P1

TRAP: Finding x-intercepts when the question asks where two graphs meet.

FIX: For intersections, set the two equations equal, not equal to zero unless one graph is an axis.

25 Expanded circle centre sign error

P1 HIGH COST

TRAP: Reading $x^2 + y^2 + 2gx + 2fy + c = 0$ as centre (g, f) .

FIX: Centre is $(-g, -f)$. Complete the square to confirm.

26 Tangent/radius gradient edge-case error

P1

TRAP: Using the same gradient for tangent and radius, or missing vertical/horizontal cases.

FIX: Normally $m_{\text{tan}} = -1/m_{\text{rad}}$. If radius is horizontal, tangent is vertical; if radius is vertical, tangent is horizontal.

27 Line gradient formula reversed

P1

TRAP: Using change in x over change in y for gradient.

FIX: Gradient = change in y over change in x: $m = (y_2 - y_1)/(x_2 - x_1)$.

28 Approximate graph solution overread

P1

TRAP: Reading exact answers from a rough sketch.

FIX: Use sketches for structure and intervals. Use algebra for exact values.

29 Axis intercept confusion

P1

TRAP: Putting $x = 0$ to find the x-intercept, or $y = 0$ to find the y-intercept.

FIX: x-intercept: set $y = 0$. y-intercept: set $x = 0$.

30 Ignoring possible number of polynomial roots

P1

TRAP: Assuming a cubic has exactly three real roots.

FIX: A polynomial can have fewer real roots than its degree. Use signs, discriminants, and graph shape.

31-40. Geometry, trigonometry, radians and vectors

31 Using the wrong circle theorem

P1

TRAP: Confusing same segment, cyclic quadrilateral, and alternate segment facts.

FIX: Mark chord, tangent, centre, and cyclic quadrilateral first. Then choose the theorem.

32 Angle at centre ratio reversed

P1

TRAP: Saying the circumference angle is twice the centre angle.

FIX: The centre angle is twice the circumference angle on the same arc.

33 Similarity scale factor error

P1

TRAP: Using k for length, area, and volume.

FIX: Lengths scale by k , areas by k^2 , volumes by k^3 .

34 Sine rule ambiguous case missed

P1 HIGH COST

TRAP: Assuming SSA gives one triangle only.

FIX: When finding an angle, check θ and $180^\circ - \theta$. Keep the second only if the triangle angle sum is possible.

35 Cosine rule included-angle error

P1 HIGH COST

TRAP: Using $a^2 = b^2 + c^2 - 2bc \cos A$ when A is not between b and c .

FIX: The angle in the cosine rule is opposite the side on the left. Side a is opposite angle A .

36 Missing trig solutions in an interval

P1 HIGH COST

TRAP: Finding only the principal solution.

FIX: Use symmetry and periodicity. Test all solutions in the required interval.

37 Exact trig value mix-up

P1

TRAP: Swapping $\sin 30^\circ$ and $\cos 30^\circ$, or $\tan 60^\circ$ and $\tan 30^\circ$.

FIX: Use the 30-60-90 and 45-45-90 triangles.

38 Radian/degree confusion

P1 HIGH COST

TRAP: Using $s = r \theta$ when θ is in degrees.

FIX: Arc and sector formulae need radians: $s = r \theta$, $A = \frac{1}{2} r^2 \theta$.

39 Sector and segment mix-up

P1

TRAP: Giving sector area when the question asks for segment area.

FIX: Segment area = sector area - triangle area.

40 Bearings drawn from the wrong north line

P1

TRAP: Measuring a bearing from the destination rather than the starting point.

FIX: Draw a north line at the point the bearing is measured from, then measure clockwise.

41-50. Vectors, calculus, area, number and proportion

41 Vector parallel condition incomplete

P1

TRAP: Comparing only one component.

FIX: Parallel vectors need a common scalar multiple: $a = k b$.

42 Midpoint/vector average error

P1

TRAP: Adding coordinates without dividing by 2.

FIX: Midpoint $M = ((x_1+x_2)/2, (y_1+y_2)/2)$, or vector $m = (a + b)/2$.

43 Differentiating before simplifying

P1

TRAP: Applying calculus to a messy fraction before rewriting powers.

FIX: Expand, cancel, divide through, and rewrite as powers before differentiating.

44 Forgetting +C in indefinite integration

P1

TRAP: Losing the constant needed for $y(1) = 3$.

FIX: Write +C immediately. Use the given point or condition to find it.

45 Area vs signed integral

P1 HIGH COST

TRAP: Treating integral from a to b of $f(x) dx$ as area when the graph goes below the axis.

FIX: Find roots, split the interval, and add absolute values of each region.

46 Trapezium rule endpoint error

P1

TRAP: Doubling all ordinates.

FIX: First and last ordinates are single. Middle ordinates are doubled.

47 Direct/inverse proportion setup error

P1

TRAP: Writing $y = kx$ when y is proportional to $1/x$.

FIX: Direct: $y = kx^n$. Inverse: $y = k/x^n$.

48 Bounds and rounding treated as exact

P1

TRAP: Using rounded measurements as exact values.

FIX: Nearest unit means true value is in $[x - 0.5, x + 0.5)$.

49 Units and compound-unit conversion error

P1

TRAP: Mixing minutes and hours, cm^2 and m^2 , or density units.

FIX: Convert units before substituting into formulae.

50 Power integration at $n = -1$ mishandled

P1

TRAP: Using $x^{n+1}/(n+1)$ when $n = -1$.

FIX: The TMUA power integral rule applies for $n \neq -1$. Do not divide by zero in the formula.

51-60. Probability, statistics and core logic

51 Independence vs mutually exclusive

P1

TRAP: Thinking mutually exclusive events are automatically independent.

FIX: Independent: $P(A \text{ and } B) = P(A)P(B)$. Mutually exclusive: $P(A \text{ and } B) = 0$.

52 Conditional probability denominator error

P1 HIGH COST

TRAP: Dividing by the total sample space instead of the given condition.

FIX: $P(A|B) = P(A \text{ and } B)/P(B)$. The denominator is the "given".

53 Tree diagram without-replacement error

P1

TRAP: Keeping second-branch probabilities unchanged after an item is removed.

FIX: Without replacement, update both numerator and denominator.

54 Histogram frequency-density error

P1

TRAP: Using frequency as bar height when class widths differ.

FIX: Bar height = frequency density = frequency/class width.

55 Grouped mean midpoint error

P1

TRAP: For class $10 < x \leq 20$, using 10 or 20 instead of 15.

FIX: Estimated mean = $\text{sum}(fx)/\text{sum}(f)$, where x is the class midpoint.

56 Complement and "at least one" error

P1

TRAP: Trying to count every "at least one" case directly and missing cases.

FIX: Use complement: $P(\text{at least one}) = 1 - P(\text{none})$.

57 Confusing converse with contrapositive

P2 HIGH COST

TRAP: Treating $Q \Rightarrow P$ as equivalent to $P \Rightarrow Q$.

FIX: Only the contrapositive is equivalent: $P \Rightarrow Q$ equals not $Q \Rightarrow$ not P .

58 Necessary and sufficient backwards

P2 HIGH COST

TRAP: If $P \Rightarrow Q$, saying P is necessary.

FIX: P is sufficient for Q . Q is necessary for P .

59 "A if B" vs "A only if B" reversal

P2

TRAP: Reading both phrases as the same arrow.

FIX: "A if B" means $B \Rightarrow A$. "A only if B" means $A \Rightarrow B$.

60 Wrong negation of "if A then B"

P2 HIGH COST

TRAP: Negating $A \Rightarrow B$ as not $A \Rightarrow$ not B .

FIX: The negation is: A is true and B is false.

61-70. Logic, implication and proof traps

61 Wrong quantifier negation

P2

TRAP: Thinking “not all” means “none”.

FIX: Not all = at least one not. Not exists = none.

62 Treating “or” as exclusive

P2

TRAP: Thinking “A or B” excludes the case where both are true.

FIX: In TMUA logic, “or” is inclusive unless stated otherwise.

63 Confusing “if” with “if and only if”

P2

TRAP: Proving one direction and claiming equivalence.

FIX: Iff needs both directions: $A \Rightarrow B$ and $B \Rightarrow A$.

64 Affirming the consequent

P2 **HIGH COST**

TRAP: $P \Rightarrow Q$, Q true, therefore P true.

FIX: Invalid. Q may happen for other reasons.

65 Denying the antecedent

P2

TRAP: $P \Rightarrow Q$, P false, therefore Q false.

FIX: Invalid. The implication says nothing when P is false.

66 Misreading “must be true” vs “could be true”

P2 **HIGH COST**

TRAP: Selecting an option that is possible but not forced.

FIX: Underline must, could, may, always, never before doing maths.

67 Assuming the domain

P2

TRAP: Treating integers, positive integers, reals, and non-zero reals as the same.

FIX: Write the domain before testing examples or counterexamples.

68 “Not enough information” confused with false

P2

TRAP: If a conclusion does not follow, calling it false.

FIX: Separate “not implied” from “false”. A claim can be true but not proved by the premises.

69 Thinking multiple counterexamples are needed

P2

TRAP: Trying to disprove “for all” with many examples.

FIX: One valid counterexample disproves a universal statement.

70 Using examples to prove a universal claim

P2 **HIGH COST**

TRAP: Testing $n = 1, 2, 3$ and saying “therefore true”.

FIX: Examples suggest a conjecture. They do not prove all cases.

71-80. Proof and MCQ exam-process mistakes

71 Missing edge cases

P2 HIGH COST

TRAP: Ignoring 0, 1, -1, equality cases, fractions, and boundary values.

FIX: Before algebra, test the annoying five: 0, 1, -1, fractions, boundary.

72 Division by zero inside proof analysis

P2

TRAP: A false proof hides $a - b = 0$ in a cancellation step.

FIX: Every cancellation in a proof must have a non-zero justification.

73 Assuming $f(A)=f(B)$ means $A=B$

P2

TRAP: From $x^2 = 4$, concluding $x = 2$ only.

FIX: This only works for one-to-one functions. Many functions are not injective.

74 Proving the wrong direction

P2

TRAP: Asked to prove $P \Rightarrow Q$, proving $Q \Rightarrow P$.

FIX: Write start and finish: "Assume P. Need Q."

75 Incomplete proof by cases

P2

TRAP: Proving the even case but forgetting the odd case.

FIX: Cases must be exhaustive.

76 Proof-order sequencing error

P2 HIGH COST

TRAP: Choosing a true statement that does not follow next.

FIX: Each proof line must be justified by earlier lines, not just be true in isolation.

77 Contradiction proof starts with wrong assumption

P2

TRAP: To prove S, assuming something unrelated instead of not S.

FIX: Contradiction begins with "Assume not S" and ends by rejecting that assumption.

78 Over-solving when elimination is faster

P1 P2

TRAP: Doing a full derivation when two options can be killed by sign, size, domain, or units.

FIX: In MCQ, first ask: can I eliminate by domain, sign, units, edge case, or estimate?

79 Spending too long on one question

P1 P2 HIGH COST

TRAP: Losing easy later marks because one hard question became personal.

FIX: At 2.5-3 minutes with no route, guess, flag, and move.

80 Doing practice without tagging mistakes

P1 P2 HIGH COST

TRAP: More papers, same errors, no score movement.

FIX: After each paper, tag each wrong answer by mistake number and drill repeats.

Weekly TMUA error tracker

Print this page or copy it into a spreadsheet. The goal is not to collect mistakes; the goal is to stop repeating the same mistake number.

Week	Paper / test	Raw score	Top 3 mistake numbers	What I will drill	Fixed next week?
1					
2					
3					
4					
5					
6					

Rapid checklist by category

A Algebra and domain

- Move to one side before cancelling variables.
- Write denominator/log/root restrictions first.
- Check answers after squaring.
- For inequalities, ask "negative or possibly negative?"

B Graphs and geometry

- Inside graph transformations reverse direction.
- Circle centre signs reverse in expanded form.
- Radius and tangent are perpendicular.
- Radians are needed for arc and sector formulae.

C Calculus and data

- Simplify before differentiating/integrating.
- Area is not always the signed integral.
- Conditional probability denominator is the given event.
- Histogram height is frequency density, not frequency.

D Logic and proof

- Contrapositive is equivalent; converse is not.
- "Only if" points to the necessary condition.
- One counterexample disproves "for all".
- Examples do not prove universal statements.

Final 20-second exam protocol

Read

Underline must, could, all, some, only if.

Restrict

Domain, sign, denominator, interval.

Sketch/test

Try 0, 1, -1, boundary, rough graph.

Select

Eliminate, choose, flag, move.

Rule: if a wrong answer keeps appearing in your working, it is not a "careless mistake". It is a trained reflex. Replace it with a new reflex using the fix.